Why do marked words persist?

Corpus and experimental evidence for competition between phonological and semantic well-formedness in word formation and the lexicon

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Takeaway

A phonologically marked form survives in a language's lexicon and is tolerated in word formation when there are fewer semantically equivalent alternative forms.

Background

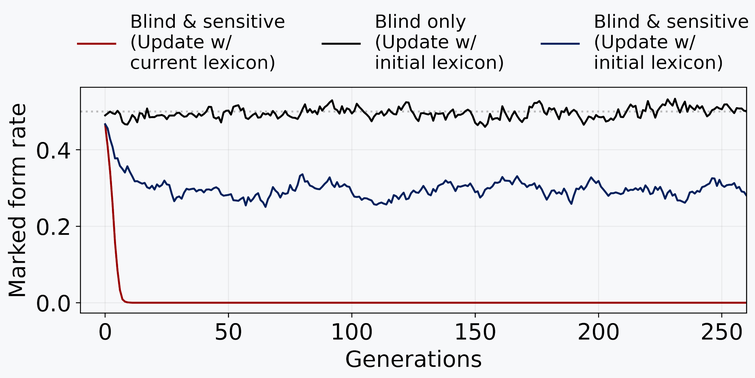
In probabilistic grammar, marked forms tend to become **marginalized** and die out over time.

Methods

MaxEnt models learned from the previous generation and produced new data for the following generation.

Table 1. Training data at initialization.

	CC	C+C
pt	100%	50%
pp	0%	50%



However, there are exceptions where marked forms are **tolerated** [1, 2, 3].

The mystery of survival: If marked forms are dispreferred, why do they survive in the lexicon?

Hypothesis

Word form flexibility (Flex) competes with phonological well-formedness.

Flex=*N* of alternative forms compatible with the target semantics.

Corpus studies

Language	Constraint	Example
English	*GEMINATE	*book.keep-er*
Finnish	*[αback][$-\alpha$back]	*kesä.loma-a* /'kesæ.lomɑ:/ "summer vacation"
Hungarian	*[αback][$-\alpha$back]	*kő.por* /'kø:.por/ "rock flour"

Data two-member compounds from Wiktionary and published datasets [4, 5]

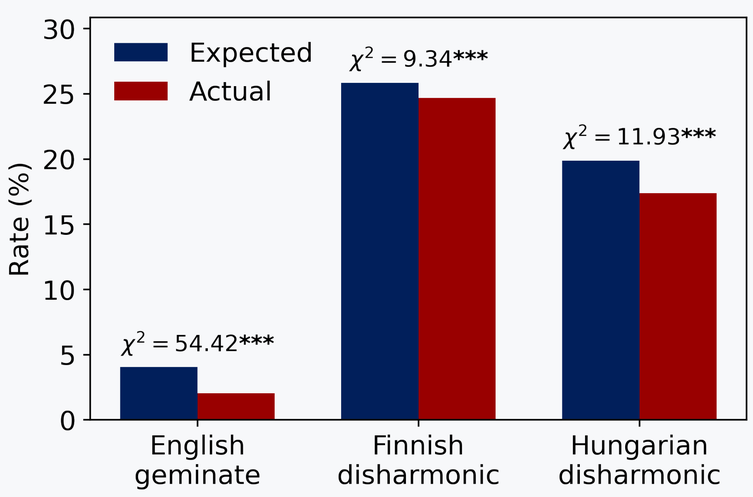
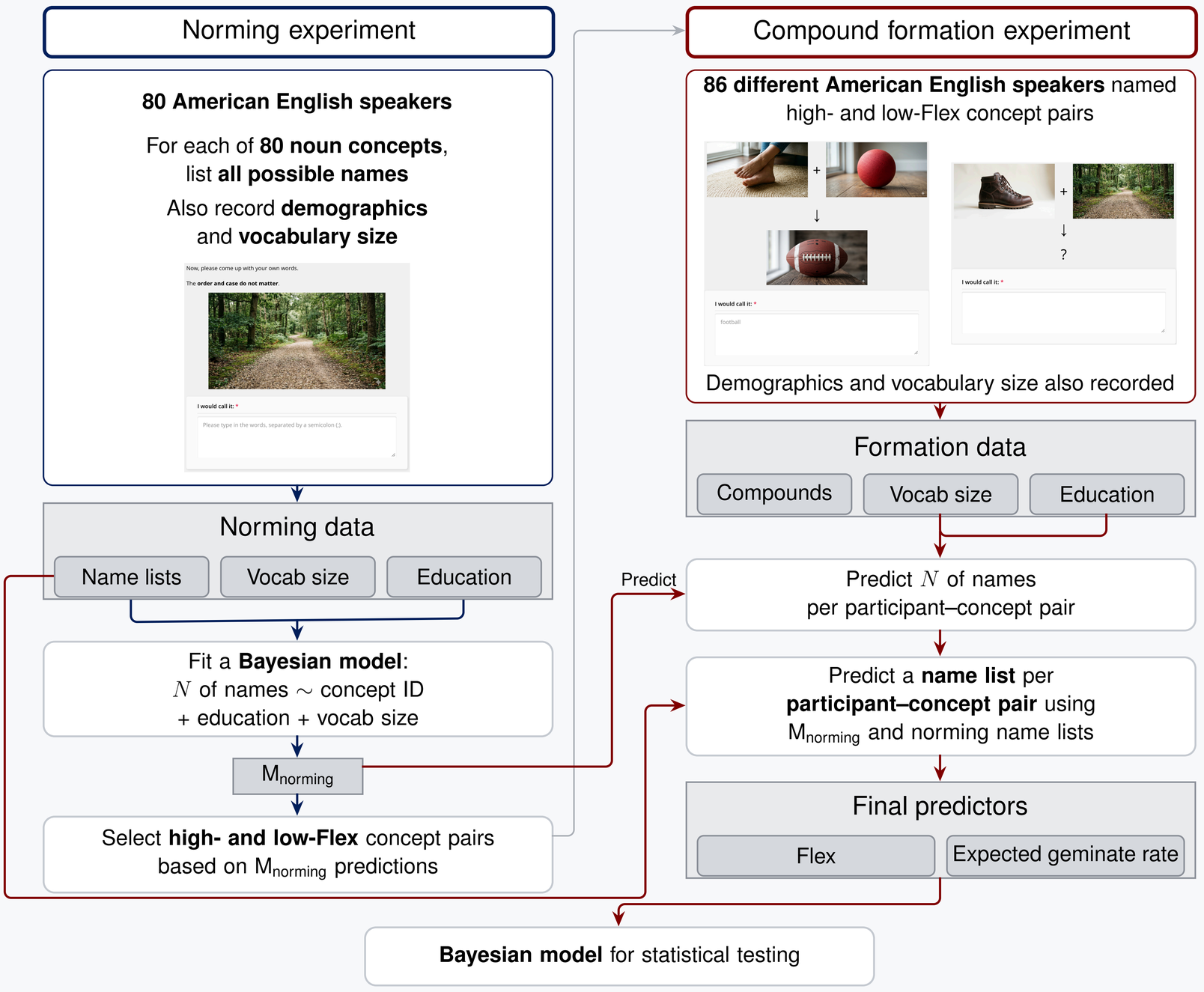


Figure 1. Geminate and disharmonic compounds are underrepresented.

Methods

- Flex(compound) = Flex(member1) $\times$ Flex(member2)
- Flex(word) = summed cosine similarity of the word's vector w/ all word vectors in the language

Compound formation experiment



Corpus results: Marked compounds tend to have fewer semantically viable candidates.

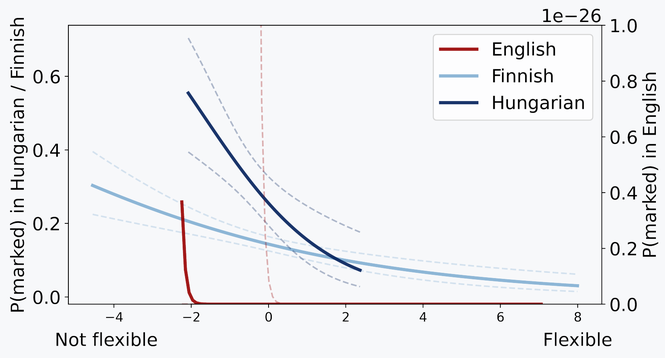


Figure 2. Flex effects on a compound's markedness.

Examining diachronic evolution through MaxEnt simulation

Flex:

For each word:

1. $k-1$ candidate forms were created, w/ the same geminate rates.
2. k was randomly sampled from a Zipfian distribution (exponent=s).

Simulation results: When the learner is constrained by flexibility in terms of the possible forms to use, marked forms survive.

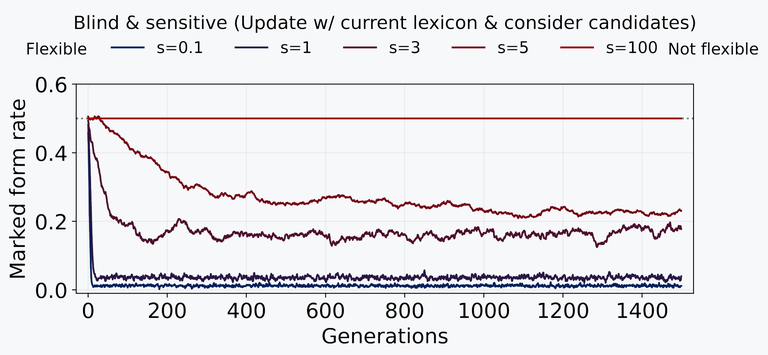


Figure 3. Simulated evolution of marked forms in the lexicon.

Experiment results: When a concept pair was less flexible, the participant was more likely to offer a geminate compound.

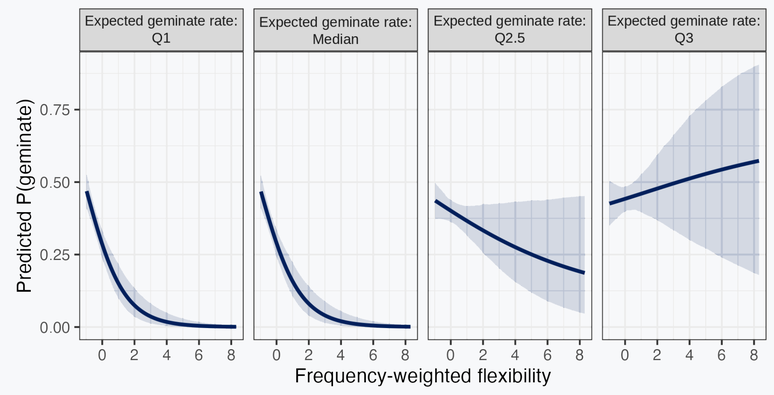


Figure 4. Model-derived interaction b/t a compound's Flex and the markedness rate of the name offered.

Selected references

[1] Kempen, G., & Harbusch, K. (2008). Comparing linguistic judgments and corpus frequencies as windows on grammatical competence: A study of argument linearization in German clauses. In A. Steube (Ed.), *The discourse potential of underspecified structures* (pp. 179–192). De Gruyter.

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[4] Gagné, C. L., Spalding, T. L., & Schmidtke, D. (2019). LADEC: The Large Database of English Compounds. *Behavior Research Methods, 51*, 2152–2179.

[5] Günther, F., & Marelli, M. (2019). Enter sandman: Compound processing and semantic transparency in a compositional perspective. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 45*(10), 1872–1882.

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